Ex:19:- [B.Q:-21]:- A is thrice effr of B and takes 10 days less to do a work than B takes. B alone in ?

Ans:- $A = 3B$; B → x days ; A = 3B → $\frac{x}{3}$ days (×3, ⅓)

$\frac{x}{3}$ days $= x - 10 \Rightarrow 3x - 30 = x$.

$\Rightarrow 3x - x = 30 \Rightarrow \boxed{x = 15 \text{ days}}$

(or)

B days assume 'x' $\Rightarrow$ effr of B $= B = \frac{1}{x}$

A effr $= 3 \times \frac{1}{x} = \frac{3}{x} \Rightarrow$ days of A $\Rightarrow A \Rightarrow \frac{x}{3}$ days.

$\therefore \frac{x}{3} = x - 10 \Rightarrow \boxed{x = 15 \text{ days}}$

Ex:-20:- [B.Q:-23]:- $A\&B = \frac{1}{72}$, $B\&C = \frac{1}{120}$, $C\&A = \frac{1}{90}$ $\Rightarrow$ A in ?

$A+B = \frac{1}{72}$, $B+C = \frac{1}{120}$, $C+A = \frac{1}{90}$ → By adding

$A+B+B+C+C+A = 2(A+B+C) = \frac{1}{72} + \frac{1}{120} + \frac{1}{90} = \frac{5+3+4}{360}$

$2(A+B+C) = \frac{12}{360} = \frac{1}{30} \Rightarrow A+B+C = \frac{1}{60}$

to find dr A days find A effr by subtracting B+C. from A+B+C.

$A = (A+B+C) - (B+C) = \frac{1}{60} - \frac{1}{120} = \frac{2-1}{120} = \frac{1}{120} = \text{effr}$

$\boxed{\text{days of A} = 120 \text{ days}}$.

Ex:-21:- [B.Q:-25]:- $A+B = \frac{1}{12}$, $B+C = \frac{1}{16}$, A worked for first 5 days and B for next 7 days, finally 'C' finishes the remaining work in 13 days. C alone in how many days?

$5A + 7B + 13C = 1 \Rightarrow 5(A+B) + 2B + 2C + 11C = 1$

$\Rightarrow 5(A+B) + 2(B+C) + 11C = 1$.

$\Rightarrow 5/12 + 2/16 + 11C = 1 \Rightarrow \frac{4\times5 + 3\times2}{48} + 11C = 1$.

$11C = 1 - \frac{26}{48} = \frac{22}{48} \Rightarrow C = \frac{2 \times 1}{48 \times 11}$ (24)

$\Rightarrow \boxed{C \text{ in } 24 \text{ days}}$

Ex.-22:- $A=\frac{1}{46}$, $B=\frac{1}{92}$ & $C=\frac{1}{23}$. A on 1st day, B on 2nd day & C on 3rd ... etc. completed the work in how many days?

Ans:- Lets consider 3 days work (1st d-A + 2nd d-B + 3rd d-C) is

$$A+B+C=\frac{1}{46}+\frac{1}{92}+\frac{1}{23}=\frac{2+1+4}{92}=\frac{7}{92}$$

lets 3 days work as a cycle. then we have to find out how many cycles can be possible by multiplying numerator (7) with as near as possible to make nearby (less) or equal to denominator (92)

$\therefore$ 13 complete cycles is possible. [$13\times7=91 \leq 92$]

$\therefore$ 13 cycles work = 13×3 days (each cycle days) work.

$$=\frac{13\times7}{92}=\frac{91}{92}=39 \text{ days work.}$$

Remaining work $=1-\frac{91}{92}=\frac{1}{92}$ of work.

the Rem work to be done by 'A'.

$\frac{1}{46}$ of work $\longrightarrow$ 1 day by A

$\frac{1}{92}$ $\longrightarrow$? $=\frac{\frac{1}{92}\times1}{\frac{1}{46}}=\frac{46}{92}=\frac{1}{2}$ day.

Total days to complete the work

$=$ 39 days (13 cycles) $+\frac{1}{2}$ day

$=39\frac{1}{2}$ days